

Lab 3: Component Modelling & Architectural Pattern Selection

Digital Campus Library Reservation Gateway

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Project Title	Digital Campus Library Reservation Gateway
Domain	Campus & Academic Operations
Key Stakeholders	Student Member, Head Librarian

1. Project Scenario

The Digital Campus Library Reservation Gateway is a self-service platform that lets students discover and reserve library resources online, while giving the Head Librarian centralized administrative control over inventory, policy, and oversight. The system must deliver a complete reservation experience across the following requirements.

Core Functionality

- Students can search the library catalog for books, study rooms, and equipment (laptops, projectors) by title, subject, or availability window.
- Students can create, modify, or cancel a reservation — a pickup slot for a book/equipment item or a time slot for a study room.
- Students receive automated confirmation, pickup-reminder, and due-date/overdue notifications tied to their reservations.
- The Head Librarian can approve or reject exception requests, manage catalog inventory (add, update, retire items), configure loan and cancellation policies, and view usage and utilization reports.
- Every user is verified through the university's central identity system before any reservation or administrative action is permitted.

Technical Constraints

- Must authenticate every request against the existing Campus Single Sign-On (SSO) / identity provider rather than maintaining a separate credential store.
- Must integrate with the university's existing Library Management System (LMS/OPAC) as the authoritative source of catalog and inventory data.
- Must deliver time-sensitive notifications (confirmations, reminders, overdue alerts) through an external email/SMS gateway.
- Must remain responsive during peak-demand periods (start of semester, exam weeks) when catalog browsing and room-booking volumes spike sharply.

2. Architectural Style Evaluation

Three candidate architectural styles were evaluated against the scenario above before an architecture was selected.

Style	Structure	Fit for this Scenario — Pros	Fit for this Scenario — Cons
Layered	Horizontal layers: presentation, business logic, data/integration	Clean separation between student portal, librarian console, and integrations; simple to build, secure, and operate for a moderate-scale campus system	Less elastic if one layer needed independent scaling far beyond the others

Style	Structure	Fit for this Scenario — Pros	Fit for this Scenario — Cons
Microservices	Small, independently deployable services	Each capability (reservations, catalog, notifications) could scale and deploy independently	Operational complexity, distributed data consistency, and network latency are disproportionate to a single-campus workload
Client-Server	Centralized server with multiple thin clients	Simple to reason about; centralized data consistency	Blurs the boundary between UI, business rules, and integrations, making it harder to isolate the SSO/LMS/notification dependencies cleanly

Layered Architecture was selected as the most appropriate style; the full justification is presented in Section 6.

3. Identified Components

Six components were identified for the Layered Architecture, distributed across the presentation, business logic, and data/integration layers.

#	Component	Layer	Responsibility
1	Student Portal	Presentation	Web/mobile self-service UI through which students search the catalog, place reservations, and track bookings.
2	Librarian Admin Console	Presentation	Staff-facing UI for the Head Librarian to manage inventory, approve exceptions, configure policy, and monitor activity.
3	Reservation Manager	Business Logic	Central orchestrator that validates, creates, modifies, and cancels reservations while enforcing loan and quota policies.
4	Campus Identity & Authentication Service	Data/Integration	Wraps the campus SSO/identity provider so every request is authenticated at one central checkpoint.
5	Catalog & Inventory Service	Data/Integration	Maintains book, room, and equipment availability by synchronizing with the existing Library Management System (LMS/OPAC).
6	Notification Service	Data/Integration	Sends booking confirmations, pickup reminders, and overdue alerts via the campus email/SMS gateway.

4. Component Interfaces

Six interfaces connect the components above, each shown in the diagram using UML ball (provided) and socket (required) notation.

Interface	Provided By	Required By	Purpose
IRequestReservation	Reservation Manager	Student Portal	Create, view, modify, and cancel a reservation for a logged-in student.
IMonitorReservations	Reservation Manager	Librarian Admin Console	Retrieve reservation activity, exceptions, and utilization data for staff oversight.

Interface	Provided By	Required By	Purpose
IVerifyIdentity	Campus Identity & Authentication Service	Reservation Manager	Authenticate the requesting user's campus credentials before any booking action.
ICheckAvailability / IUpdateInventory	Catalog & Inventory Service	Reservation Manager	Check real-time resource availability and update inventory when a reservation is made or released.
ISendNotification	Notification Service	Reservation Manager	Trigger a confirmation, reminder, or overdue alert tied to a reservation event.
IManageCatalog	Catalog & Inventory Service	Librarian Admin Console	Add, edit, retire, or adjust availability rules for catalog items.

5. UML Component Diagram

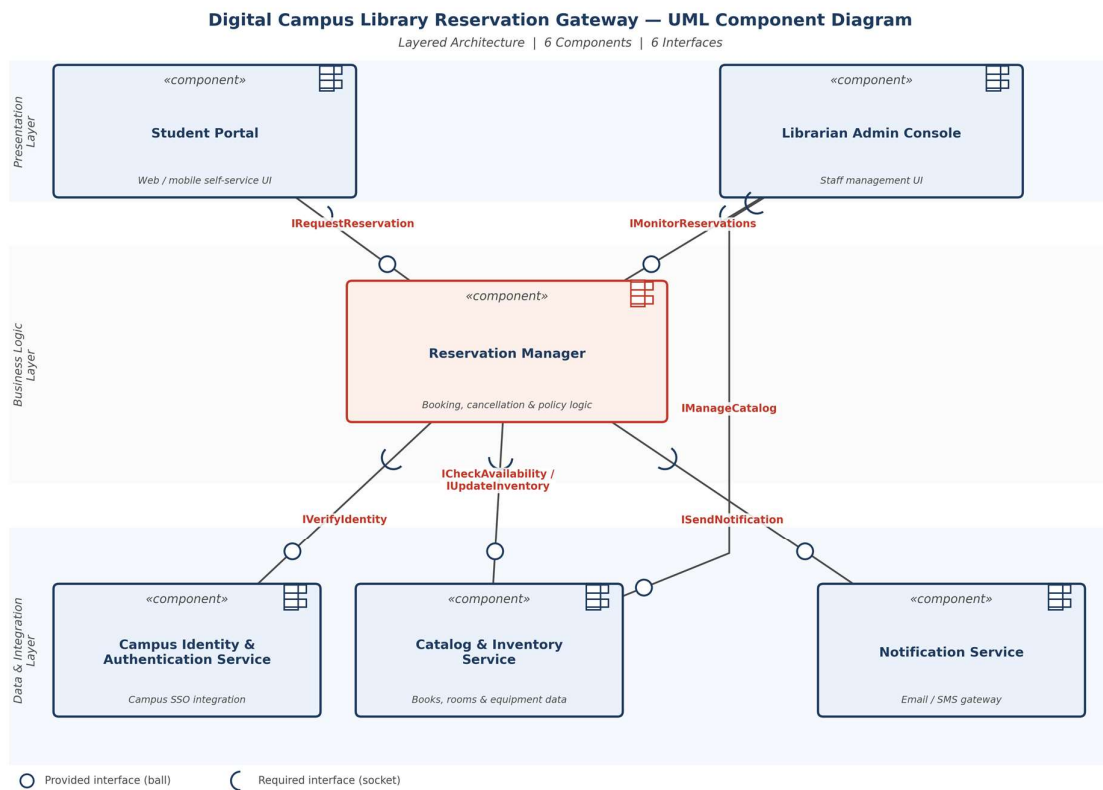


Figure 1. Component diagram for the Digital Campus Library Reservation Gateway (Layered Architecture) — 6 components and 6 interfaces shown with UML ball-and-socket notation.

6. Written Architectural Justification

Architecture Selection: We chose Layered Architecture for the Digital Campus Library Reservation Gateway.

Reason 1 — Independent evolution of two front ends

The Gateway's value lies in coordinating three fairly stable concerns: a student-facing booking experience, librarian administrative control, and integration with pre-existing campus systems. A layered structure cleanly separates the presentation layer (Student Portal, Librarian Admin Console) from the business logic layer (Reservation Manager) and the data/integration layer (Authentication, Catalog, and Notification services). This lets the two front ends evolve independently — for example, redesigning the student mobile app during a future upgrade — without touching the reservation and policy rules that both interfaces depend on.

Reason 2 — Operational complexity matched to scenario scale

The Gateway does not need to scale each capability independently the way a large e-commerce platform would. The operational overhead of a microservices architecture — independent deployments, service discovery, distributed data consistency — would be disproportionate to a single-campus workload, whereas the simpler, centrally deployed layered style matches the campus IT team's staffing and support model, while still isolating integration points (SSO, LMS, notification gateway) in their own layer for easy replacement if the university changes vendors.

Security Advantage

Every reservation request is funnelled through the Reservation Manager, which must call the Campus Identity & Authentication Service's IVerifyIdentity interface before any booking, cancellation, or catalog change is processed. This creates one consistent authorization checkpoint and audit trail for both students and library staff, rather than each presentation-layer client re-implementing its own credential checks — significantly reducing the risk of an unauthenticated or spoofed request reaching the catalog or inventory data.

Performance Benefit

Catalog and availability look-ups — the most frequent, read-heavy operation, especially during the start of semester or exam-week room-booking rushes — are isolated in the Catalog & Inventory Service at the data/integration layer. This lets the Reservation Manager cache frequently requested availability data at that boundary without touching the presentation or business logic layers, reducing repeated round-trips to the university's Library Management System and keeping search and browse response times low even under peak load.